Numerical Metrics

Split	ROC-AUC	Accuracy	Balanced Accuracy	Sensitivity / Recall	Specificity	Precision	F1-score	Brier Score	PR-AUC	TN	FP	FN	TP	Expected Calibration Error / ECE
validation	0.5335	0.7013	0.5571	0.7391	0.3750	0.9107	0.8160	0.2987	0.9045	3	5	18	51	0.2987
test	0.5902	0.6795	0.5773	0.7101	0.4444	0.9074	0.7967	0.3205	0.9024	4	5	20	49	0.3205

COMBINED BINARY CLASSIFICATION

- Validation ROC-AUC is 0.591, the best among the three validation results, but still weak.

- Test ROC-AUC is 0.471, worse than PET and below random ranking.

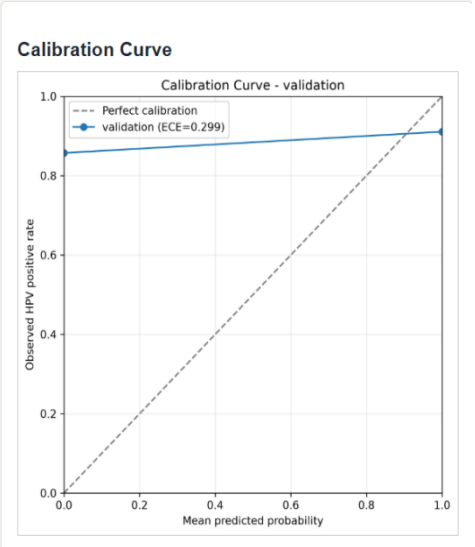
- Validation balanced accuracy is 0.589, better than clinical and PET, but test balanced accuracy drops to 0.471.

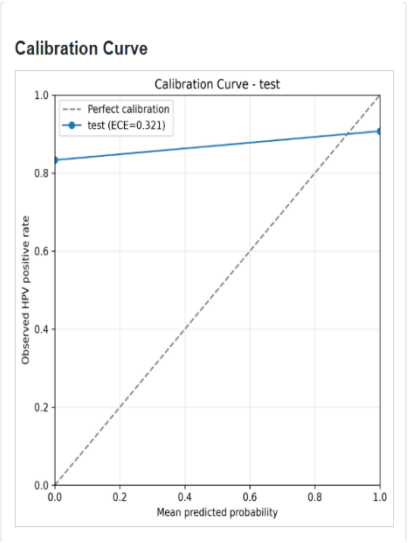
- Accuracy and F1-score look high because the dataset is dominated by HPV-positive patients.

- Specificity remains the major weakness. It is 0.250 on validation and 0.000 on test.

- PR-AUC is high on validation, but because the positive rate is already very high, this does not prove strong discrimination.

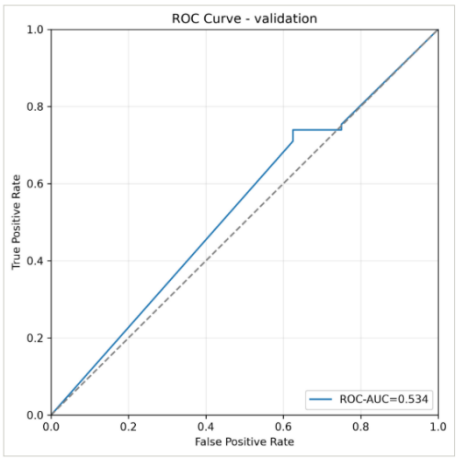
- Brier score and ECE are better than PET but worse than an ideal calibrated model.



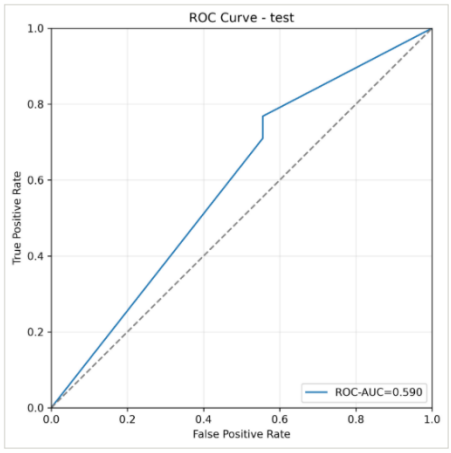


The combined calibration curve shows overconfident predictions. In validation, the [0.0, 0.1) bin has an observed HPV-positive rate of 0.714, which means many patients predicted as very low probability are actually positive. In test, the [0.0, 0.1) bin has an observed HPV-positive rate of 1.000. The high-probability bin also contains some HPV-negative patients. This is why ECE remains high and why the probabilities are not trustworthy.

ROC Curve

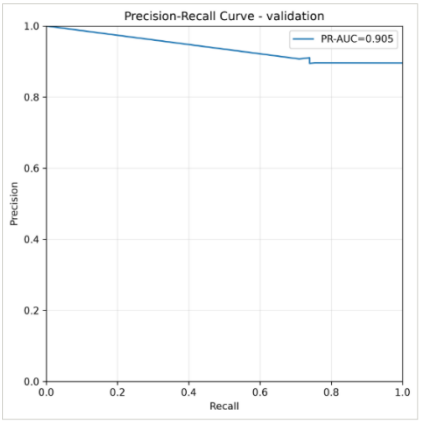


ROC Curve

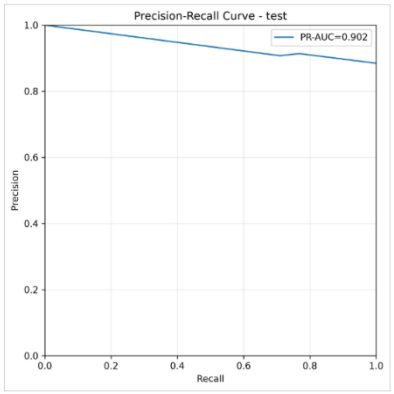


The validation ROC curve is the best among the three experiments, with ROC-AUC 0.591, but this is still weak. The test ROC-AUC drops to 0.471, which is below random ranking. Ideally, the combined ROC curve should be clearly above both the clinical and PET curves on both validation and test. Instead, the val]\

Precision-Recall Curve



Precision-Recall Curve

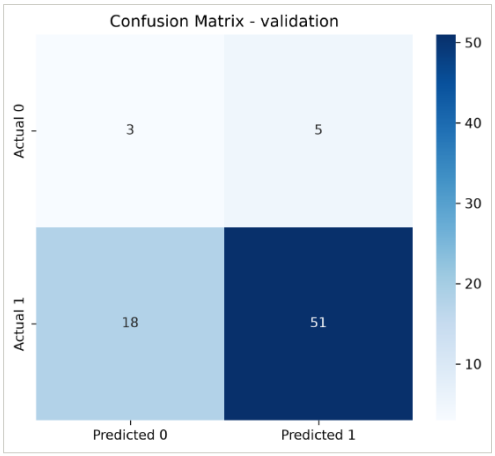


The validation PR-AUC is 0.913, but the positive class is already very common.

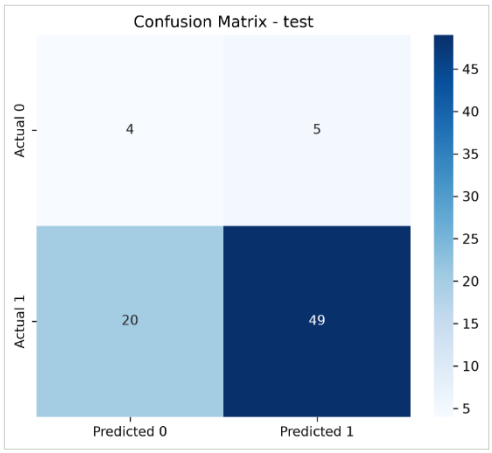
The test PR-AUC is 0.879, close to the test positive prevalence. This means the PR curve does not prove strong discrimination. It mainly reflects the fact that most patients are HPV-positive.

idation curve shows only a small improvement, and the test curve does not generalize.

Confusion Matrix Heatmap

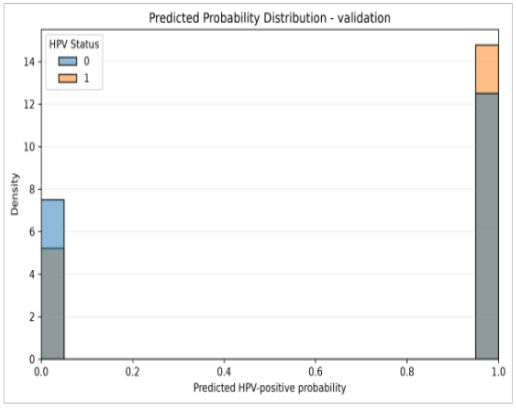


Confusion Matrix Heatmap

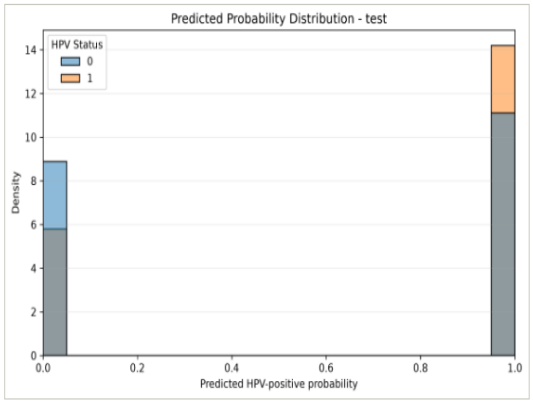


The validation heatmap looks slightly better than clinical and PET because it has 2 true negatives and only 5 false negatives. However, the test heatmap has 0 true negatives, 9 false positives, 4 false negatives, and 65 true positives. This means the model did not keep its validation improvement on the independent test set.

Probability Distribution

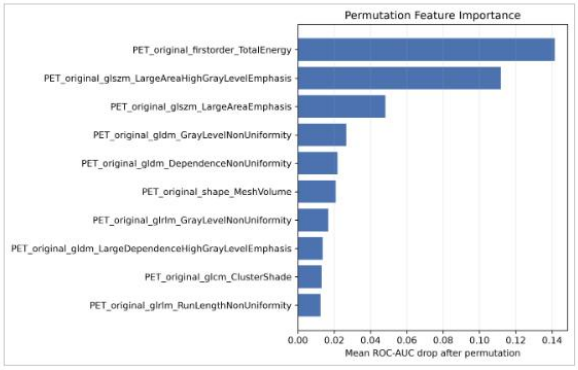


Probability Distribution

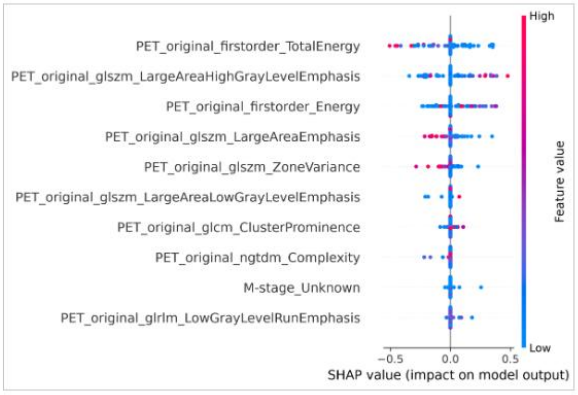


The ideal combined probability distribution would show HPV-negative cases near 0 and HPV-positive cases near 1. The combined plot instead shows overconfident predictions, with many patients pushed toward very high probability and some positives pushed toward very low probability. This explains why accuracy can look acceptable while calibration and ROC-AUC remain poor.

Feature Importance Plot



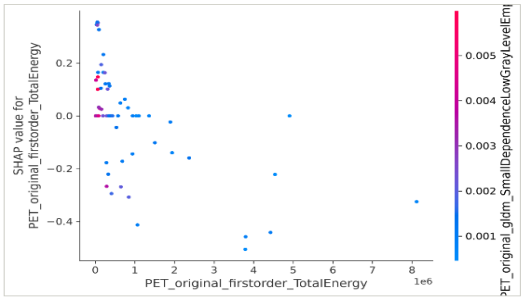
SHAP Summary Plot



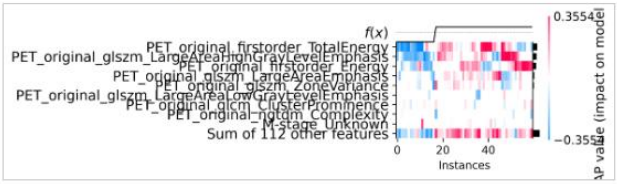
The permutation importance plot is dominated by PET features, especially LargeAreaHighGrayLevelEmphasis, VoxelVolume, LargeAreaEmphasis, DependenceNonUniformity, and MeshVolume. Clinical variables do not appear among the dominant features. This suggests that adding clinical features did not strongly change what the MLP relied on.

The SHAP summary plot also shows PET features at the top, especially TotalEnergy, Energy, LargeAreaEmphasis, and LargeAreaHighGrayLevelEmphasis. This means the combined model is mostly acting like a PET-driven model, with clinical variables contributing less.

SHAP Dependence Plot



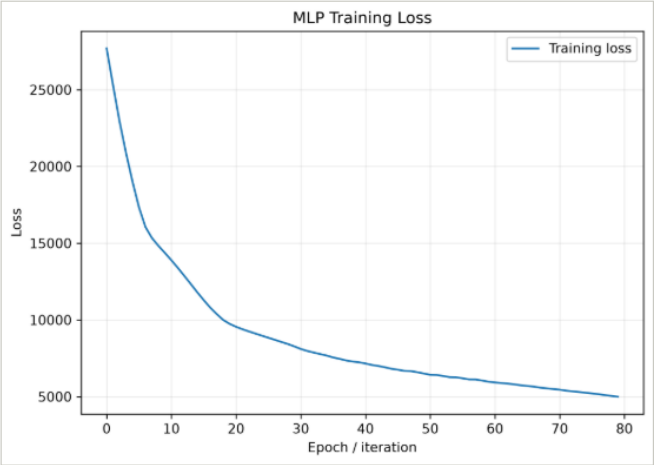
SHAP Heatmap Plot



The dependence plot automatically selected PET_original_firstorder_TotalEnergy. This matches the PET-only model and confirms that the combined model is driven by similar radiomics intensity/energy patterns. The dependence plot does not show that clinical variables rescued the prediction problem.

The SHAP heatmap should ideally show a consistent explanatory pattern separating positive and negative patients. In the combined model, the heatmap is dominated by PET radiomics effects. This supports the conclusion that the model did not learn a balanced clinical-plus-PET decision structure.

Training Loss Curve



The training loss curve shows that the network was able to train, but the test ROC-AUC and specificity show poor generalization. With 121 features and only 359 training patients, the model can learn training patterns that are not stable on test data.